General Solution Scaling Relations in Linear Elastic Dynamic Rupture Models

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Key Points:

- We prove the generality of solution scaling in linear elastic dynamic rupture problems with boundary integral equation analysis.
- We derive solution scaling relations for various commonly used friction frameworks.
- Our results help earthquake modelers design simulations more efficiently and better interpret their results.

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Abstract

Dynamic rupture models have been widely used to enhance our understanding of earthquake processes and associated hazards. A commonly held insight among modelers is that if the initial stress and fault rheology parameters are scaled by a certain factor in a specific manner, the solution will scale accordingly. While this principle has been applied in numerical simulations, particularly to optimize parameter studies and interpret results through non-dimensionalization, its general applicability to more complex modeling scenarios has not been rigorously established. In this study, we provide a proof for the generality of solution scaling in linear elastic dynamic rupture problems using a boundary integral equation formulation. The proof holds under arbitrary complexity in fault geometry, medium velocity structure, and fault rheology parameters, provided the uniqueness of the solution is satisfied. We consider four types of fault rheology: linear slip-weakening, rate-and-state friction, linear time-weakening, and general composite rheology with different slip rate dependencies at different slip rates. For each rheology, we demonstrate how to scale the stress and rheology parameters such that the resulting rupture kinematics, including slip, slip rate, and traction evolution, scale accordingly. These results show that solution scaling is a general property of linear elastic dynamic rupture physics. This finding has practical implications for the design and interpretation of large-scale simulation campaigns and provides theoretical support for widely used scaling practices in the dynamic rupture modeling community, contributing a rigorous mathematical basis for their continued use.

Plain Language Summary

Dynamic rupture models describe how earthquakes start and propagate along faults in the Earth's crust. These models are based on physical laws governing the forces and friction acting on faults and are typically solved using large-scale numerical simulations. They are widely used to study the physical mechanisms of earthquakes and to assess seismic hazards in earthquake-prone regions. A common empirical practice among earthquake modelers is to scale input parameters—such as stress levels or fault friction properties—by a constant factor, with the assumption that the outputs of the simulation will scale proportionally. This practice simplifies model design and facilitates comparisons across different parameter sets. However, until now, it had not been rigorously demonstrated whether this scaling behavior truly holds beyond the simplest cases. In this study,

we provide a mathematical proof showing that this scaling principle applies broadly in models that assume linear elasticity and certain widely used fault boundary conditions. This includes faults with complex geometries and heterogeneous material properties. Our results offer a theoretical foundation for a widely used modeling strategy, enabling more efficient parameter exploration, improved interpretation of simulation results, and better-informed dynamic rupture scenario modeling.

1 Introduction

Dynamic rupture models of earthquakes describe how slip and rupture evolve spontaneously, given fault geometry, the velocity structure of the surrounding medium, initial stress, and faulting rheology (e.g., friction law, viscous flow law) (Harris et al., 2018; Ramos et al., 2022). In recent decades, these models have been widely applied to enhance our understanding of earthquake processes and associated hazards, including exploring the mechanical controls on rupture dynamics and the resulting ground motions (e.g., Andrews, 1976; Das & Aki, 1977; Day, 1982; Harris & Day, 1993; Oglesby et al., 1998; Kame et al., 2003; Bizzarri & Cocco, 2003; Oglesby, 2005; Ando & Yamashita, 2007; Ma & Beroza, 2008; Dalguer & Day, 2009; Kaneko & Lapusta, 2010; Lozos et al., 2011; Gabriel et al., 2013; Shi & Day, 2013; Huang et al., 2014; J. Xu et al., 2015; Withers et al., 2018; Gabriel et al., 2024), explaining the rupture dynamics of past earthquakes (e.g., Olsen et al., 1997; Oglesby & Day, 2001; Aochi & Fukuyama, 2002; Dunham & Archuleta, 2004; Pulido & Dalguer, 2009; Lozos, 2016; Kyriakopoulos et al., 2017; Ando & Kaneko, 2018; Ulrich et al., 2019; Gallovič et al., 2019; Lozos & Harris, 2020; Cortez et al., 2021; Taufiqurrahman et al., 2023; Jia et al., 2023; Ding et al., 2023; Z. Wang et al., 2023), and developing hypothetical scenarios for physics-based seismic hazard analysis (e.g., Hok et al., 2011; Oglesby & Mai, 2012; Ryan et al., 2015; Liu & Duan, 2018; Kyriakopoulos et al., 2019; Yao & Yang, 2022; Tarnowski et al., 2022; Ulrich et al., 2022; Z. Wang et al., 2022; Li et al., 2023).

Numerical simulations are typically required to find the solution to a dynamic rupture problem because the model's complexities render an analytical solution impractical. A common piece of wisdom among dynamic rupture modelers is that if one scales the stress and fault rheology parameters by a numerical factor in a particular manner, the solution will be scaled by the same numerical factor. For example, consider that we obtain a dynamic rupture simulation using linear slip-weakening fault rheology. If one

scales up the initial normal and shear stress by a factor of ten, and at the same time scales up the slip-weakening distance by a factor of ten, while keeping all other parameters unchanged, the new simulation will have the same spatial-temporal evolution of rupture, except that the slip, slip rate, and slip acceleration will be ten times greater than in the initial model.

Such solution scaling relations might seem self-evident from dimensional analysis. They have been widely applied in dynamic rupture modeling research during the exploration stage to optimize parameter search, as well as in the interpretation stage to non-dimensionalize the simulation results (e.g., Day, 1982; Fukuyama & Madariaga, 1998). However, rigorous proof has only been shown for a planar fault in a homogeneous medium (e.g., Fukuyama & Madariaga, 1998). It is not immediately obvious that solution scaling relations can be extended to cases where the fault geometry, the medium's velocity structure, or fault rheology is more complex, even though they empirically appear to always hold for elastic models.

In this study, we address the generality of solution scaling in linear elastic dynamic rupture models. Such a dynamic rupture problem can be formalized as solving for the slip and traction evolutions on the internal boundaries (i.e., the fault) using a set of general equations, which include three boundary integral equations that describe the elastodynamic interactions, along with several fault rheology equations that describe the traction-slip relation on the internal boundaries. We investigate the behavior of solution scaling under several commonly used fault rheologies: linear slip-weakening, rate-and-state friction, linear time-weakening, and general composite rheology with different slip-rate dependencies at different slip rates. We show that if one obtains a solution for a dynamic rupture problem with a particular set of parameters, one may scale the solution by some numerical factor, and the scaled solution will satisfy a new dynamic rupture problem, where certain stress and fault rheology parameters are scaled by the same numerical factor. Assuming uniqueness of the solution, the scaled solution is the correct solution to the scaled problem, and the solution scaling relation is proved. Since our derivation approach is general, the solution scaling relations should hold for arbitrarily complex configurations under linear elasticity.

2 Formulation of a dynamic rupture problem as a boundary integral equation problem

A dynamic rupture problem in a purely elastic medium can be formulated as finding the solution to a set of integral equations (Tada, 2009). The fault Γ can be treated as an internal surface embedded in the medium. The representation theorem indicates that the displacement field can be expressed as a surface integral (e.g., Aki & Richards, 2002),

$$u_k(\boldsymbol{x},t) = \int_\Gamma \mathrm{d}S(\boldsymbol{\xi}) \int_0^t \mathrm{d}\tau \Delta u_i(\boldsymbol{\xi},\tau) c_{ijpq} n_j(\boldsymbol{\xi}) \frac{\partial}{\partial \xi_q} G_{kp}(\boldsymbol{x}, t-\tau; \boldsymbol{\xi}, 0), \tag{1}$$

in which $u_k(\boldsymbol{x},t)$ is the k-th component of displacement at location $\boldsymbol{x}$ and time t. $\boldsymbol{\xi}$ is a location variable on the fault. $\mathrm{d}S(\boldsymbol{\xi})$ is the fault surface area at $\boldsymbol{\xi}$. $\Delta u_i(\boldsymbol{\xi},\tau)$ is the i-th component of slip at location $\boldsymbol{\xi}$ and time τ. c_{ijpq} is the generalized fourth-order tensor that describes the stress-strain relation. $n_j(\boldsymbol{\xi})$ is the unit normal vector on the fault at $\boldsymbol{\xi}$. $G_{kp}(\boldsymbol{x}, t-\tau; \boldsymbol{\xi}, 0)$ is the k-th component of the displacement response at location $\boldsymbol{x}$ and time $t-\tau$, produced by a unit impulse force along the p-th axis applied at location $\boldsymbol{\xi}$ and time 0, which is usually referred to as the elastodynamic Green's function. The Green's function is the same as long as the medium properties remain unchanged and the boundary conditions are homogeneous (e.g., zero displacement or traction).

Denoting $\sigma_{kl}(\boldsymbol{x},t)$ as the stress tensor field, the elastic constitutive relation yields,

$$\sigma_{kl}(\boldsymbol{x},t) = \sigma^0_{kl} + c_{klrs} \frac{\partial}{\partial x_s} u_r(\boldsymbol{x},t), \tag{2}$$

in which σ^0_{kl} is the initial stress tensor. We may then substitute Equation (1) into (2) and place the observation point $\boldsymbol{x}$ on the fault. With some necessary regularization, we can express the traction $T_i(\boldsymbol{x},t)$ on the fault as the initial traction $T^0_i(\boldsymbol{x})$ plus a surface integral that contains the slip rate function $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},\tau)$,

$$T_i(\boldsymbol{x},t) = T^0_i(\boldsymbol{x}) + \int_\Gamma \mathrm{d}S(\boldsymbol{\xi}) \int_0^t \mathrm{d}\tau \hat{K}_{ij}(\boldsymbol{x}, t-\tau; \boldsymbol{\xi}, 0) \Delta\dot{u}_j(\boldsymbol{\xi},\tau) \quad (x \in \Gamma), \tag{3}$$

in which $\hat{K}_{ij}$ is the integral kernel after regularization, which includes some differential operators on the slip rate. Depending on the specific medium setup, such as full space, half space, or layered velocity structure, $\hat{K}_{ij}$ can take different forms. Although obtaining the specific mathematical expression for $\hat{K}_{ij}$ is non-trivial, $\hat{K}_{ij}$ should always exist and can therefore be treated as a given condition.

The goal in a dynamic rupture problem is to find the solution for $T_i(\boldsymbol{x},t)$ and $\Delta\dot{u}_i(\boldsymbol{\xi},t)$ on the fault. After the solution is found, the displacement field can be obtained using

Equation (1). In three dimensions, since each vector has three components, there are six unknown variables in total. However, Equation (3) only provides three equations, so we still need three more equations to constrain the solution.

Let the two shear traction directions and the normal traction direction be denoted as $\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})$, $\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})$, and $\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})$, respectively. An obvious condition is that the slip rate vector $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$ must be parallel to the fault,

$$\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi}) = 0, \tag{4}$$

where $\cdot$ denotes the dot product.

We still need two more equations. These will come from the fault boundary conditions, often referred to as fault rheology or the fault friction law, which relates the fault normal traction to the shear traction through a friction coefficient μ. Since there are two components of shear traction given the fault normal vector, the fault boundary conditions can provide two equations. We assume that the shear traction vector must have the same direction as the fault slip rate $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$,

$$\boldsymbol{T}(\boldsymbol{\xi}, t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi}, t)\left(\boldsymbol{T}(\boldsymbol{\xi}, t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\right)\left(\frac{\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)}{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)|} \cdot \hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})\right) \tag{5}$$

$$\boldsymbol{T}(\boldsymbol{\xi}, t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi}, t)\left(\boldsymbol{T}(\boldsymbol{\xi}, t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\right)\left(\frac{\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)}{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)|} \cdot \hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})\right) \tag{6}$$

We still need to know the expression for $\mu(\boldsymbol{\xi}, t)$ to solve for $T_i(\boldsymbol{x}, t)$ and $\Delta\dot{u}_i(\boldsymbol{\xi}, t)$ on the fault. $\mu(\boldsymbol{\xi}, t)$ depends on the specific formulation of fault rheology, such as linear slip-weakening or the empirical rate-and-state friction law. $\mu(\boldsymbol{\xi}, t)$ itself is often described by a set of ordinary differential equations, as in the empirical rate-and-state friction law, with additional state variables and evolution laws. Mathematically, since the formulation for $\mu(\boldsymbol{\xi}, t)$ can be complicated and nonlinear, it is not guaranteed that a solution exists or that the solution is unique.

Nevertheless, in this paper we assume that, for the fault rheology formulations we explore, there is always a solution for $T_i(\boldsymbol{x}, t)$ and $\Delta\dot{u}_i(\boldsymbol{\xi}, t)$ on the fault, and that the solution is unique, without rigorously proving it. This assumption of uniqueness will be key to the analysis in this paper. While it may be possible to mathematically prove the existence and uniqueness of the solution, such a proof is beyond the scope of this study.

Physically, it is reasonable to believe that there should be a unique solution to a real-world problem. In practice, the majority of dynamic rupture simulations in the literature implicitly assume that models operate in the parameter domain where the solution is unique and stable. With all that said, we note that we are taking a leap of faith in assuming the existence and uniqueness of the solution, and there may be extreme cases where the following analysis does not apply.

3 Proving solution scaling relation in an arbitrary condition

Up until this point, we have deduced six equations (the three equations in (3), and Equations (4), (5), (6)) to solve for the six components of the unknown variables (three for $\boldsymbol{T}(\boldsymbol{\xi}, t)$ and three for $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$) with a given fault rheology $\mu(\boldsymbol{\xi}, t)$. The first three equations are commonly referred to as boundary integral equations (Tada, 2009). Solving these equations under specific conditions can be non-trivial, and a numerical approach is often required. However, for the purpose of proving the scaling relations between solution and parameters, we do not need to know the exact expression for the solution. In the following section, we discuss two commonly used fault rheologies in dynamic rupture research: linear slip-weakening and rate-and-state friction laws.

3.1 Linear slip-weakening

In a linear slip-weakening friction framework, a location on the fault can have two types of boundary conditions. Before the shear traction reaches the yielding strength, slip remains zero. After the shear traction exceeds the yielding strength, the traction-slip relation follows a linear slip-weakening function (e.g., Ida, 1972; Palmer et al., 1973; Andrews, 1976; Day, 1982). The problem has two types of boundary condition, and the partitioning of the domain changes dynamically over time.

We denote the subset of the fault that has ruptured at time t as $\Gamma_A(t)$, and the complementary subset that has not ruptured as $\Gamma_B(t)$, and they can be defined as,

$$\boldsymbol{\xi} \in \begin{cases} \Gamma_A(t), \text{ if } |\boldsymbol{T}(\boldsymbol{\xi}, t) - \big(\boldsymbol{T}(\boldsymbol{\xi}, t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\big)\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| = \mu_y(\boldsymbol{\xi})|\boldsymbol{T}(\boldsymbol{\xi}, t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| \text{ or } |\Delta\boldsymbol{u}(\boldsymbol{\xi}, \tau)| > 0 \\ \Gamma_B(t), \text{ otherwise} \end{cases} \tag{7}$$

where $\mu_y(\boldsymbol{\xi})$ is the static friction coefficient.

To describe such a hybrid boundary condition problem, we need to revise Equations (5) and (6) to include the condition before the shear traction reaches the yielding strength,

$$\begin{cases} \boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\Big(\dfrac{\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})\Big), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ \Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{8}$$

$$\begin{cases} \boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\Big(\dfrac{\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})\Big), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ \Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{9}$$

After rupturing, the friction coefficient μ drops linearly from μ_y to μ_d over a critical slip weakening distance D_0,

$$\mu(\boldsymbol{\xi},t) = \begin{cases} \mu_y(\boldsymbol{\xi})\Big(1-\dfrac{\int_0^t|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau}{D_0(\boldsymbol{\xi})}\Big)+\mu_d(\boldsymbol{\xi}), & \displaystyle\int_0^t|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau < D_0(\boldsymbol{\xi}) \\ \mu_d(\boldsymbol{\xi}), & \displaystyle\int_0^t|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau \geq D_0(\boldsymbol{\xi}) \end{cases} \quad (\boldsymbol{\xi}\in\Gamma_A(t)), \tag{10}$$

Once we substitute Equation (10) into (8) and (9), Equation (8) and (9) would only be about fault traction $\boldsymbol{T}(\boldsymbol{\xi},t)$ and fault slip rate $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)$.

We want to prove the following statement: In a linear slip-weakening framework, considering a general configuration, let's say we have obtained the solution under the conditions of a given initial fault traction $\boldsymbol{T^0}$, yielding friction coefficient μ_y, dynamic friction coefficient μ_d, and slip-weakening distance D_0. If we linearly scale $\boldsymbol{T^0}$ and D_0 by a factor of C, while keeping the other parameters unchanged, the solutions $\boldsymbol{T}(\boldsymbol{\xi},t)$ and $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)$ will also be scaled by a factor of C.

We denote the solutions that satisfy the condition before 'scaling' as $\boldsymbol{T^a}(\boldsymbol{\xi},t)$ and $\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$. We can substitute these solutions into Equations (3), (4), (7), (8), (9), and (10),

$$T_i^a(\boldsymbol{x},t) = T_i^0(\boldsymbol{x}) + \int_\Gamma \mathrm{d}S(\boldsymbol{\xi})\int_0^t \mathrm{d}\tau\hat{K}_{ij}(\boldsymbol{x},t-\tau;\boldsymbol{\xi},0)\Delta\dot{u}_j^a(\boldsymbol{\xi},\tau) \quad (x\in\Gamma) \tag{11}$$

$$\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi}) = 0 \tag{12}$$

$$\boldsymbol{\xi} \in \begin{cases} \Gamma_A(t), \text{ if } |\boldsymbol{T^a}(\boldsymbol{\xi},t) - \big(\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\big)\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| = \mu_y(\boldsymbol{\xi})|\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| \text{ or } |\Delta\boldsymbol{u^a}(\boldsymbol{\xi},\tau)| > 0 \\ \Gamma_B(t), \text{ otherwise} \end{cases} \tag{13}$$

$$\begin{cases} \boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\Big(\dfrac{\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})\Big), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ \Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{14}$$

$$\begin{cases} \boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\Big(\dfrac{\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})\Big), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ \Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{15}$$

$$\mu(\boldsymbol{\xi},t) = \begin{cases} \mu_y(\boldsymbol{\xi})\Big(1 - \dfrac{\int_0^t|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau}{D_0(\boldsymbol{\xi})}\Big) + \mu_d(\boldsymbol{\xi}), & \displaystyle\int_0^t|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau < D_0(\boldsymbol{\xi}) \\ \mu_d(\boldsymbol{\xi}), & \displaystyle\int_0^t|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau \geq D_0(\boldsymbol{\xi}) \end{cases} \quad (\boldsymbol{\xi}\in\Gamma_A(t)), \tag{16}$$

If we scale $\boldsymbol{T^0}$, and D_0 all by a factor of C, to prove the aforementioned statement, we need to show that $C\boldsymbol{T^a}(\boldsymbol{\xi},t)$ and $C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$ satisfy the new set of equations. We can simply substitute the proposed solutions into the equations and check,

$$CT_i^a(\boldsymbol{x},t) = CT_i^0(\boldsymbol{x}) + \int_\Gamma \mathrm{d}S(\boldsymbol{\xi})\int_0^t \mathrm{d}\tau\hat{K}_{ij}(\boldsymbol{x},t-\tau;\boldsymbol{\xi},0)C\Delta\dot{u}_j^a(\boldsymbol{\xi},\tau) \quad (x\in\Gamma) \tag{17}$$

$$C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi}) = 0 \tag{18}$$

$$\boldsymbol{\xi} \in \begin{cases} \Gamma_A(t), \text{ if } |C\boldsymbol{T^a}(\boldsymbol{\xi},t) - \big(C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\big)\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| = \mu_y(\boldsymbol{\xi})|C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| \text{ or } |C\Delta\boldsymbol{u^a}(\boldsymbol{\xi},\tau)| > 0 \\ \Gamma_B(t), \text{ otherwise} \end{cases} \tag{19}$$

$$\begin{cases} C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\Big(\dfrac{C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})\Big), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{20}$$

$$\begin{cases} C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\left(\dfrac{C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})\right), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{21}$$

$$\mu(\boldsymbol{\xi},t) = \begin{cases} \mu_y(\boldsymbol{\xi})\left(1-\dfrac{\int_0^t|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau}{CD_0(\boldsymbol{\xi})}\right)+\mu_d(\boldsymbol{\xi}) & , \int_0^t|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau < CD_0(\boldsymbol{\xi}) \\ \mu_d(\boldsymbol{\xi}) & , \int_0^t|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},\tau)|\,\mathrm{d}\tau \geq CD_0(\boldsymbol{\xi}) \end{cases} \quad (\boldsymbol{\xi}\in\Gamma_A(t)), \tag{22}$$

One can find that all the scaling factors C in the equations will cancel out, and Equations (17) to (22) are identical to Equations (11) to (16). This means that the solutions $C\boldsymbol{T^a}(\boldsymbol{\xi},t)$ and $C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$ satisfy the new condition. Assuming the uniqueness of the solution, if $C\boldsymbol{T^a}(\boldsymbol{\xi},t)$ and $C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$ happen to be a solution, they are the unique solution. Since our proof does not specify the model configuration, such as the shape of the free surface, the medium velocity structure, or fault geometry, the conclusion should be general and applicable to all such dynamic rupture problems with linear slip-weakening fault rheology.

3.2 Rate-and-state friction

We can apply a similar analysis as in Section 3.1 to explore the solution scaling for dynamic rupture models that adopt a rate-and-state type friction law as the fault boundary condition, where the friction coefficient μ depends not only on the slip rate $\Delta\dot{\boldsymbol{u}}$, but also on a state variable θ that captures the effects of slip history. The state variable θ is a new variable to solve for, and its evolution is described by an ordinary differential equation, often referred to as the state evolution law.

Here we investigate the classic empirical formulation of rate-and-state friction (e.g., Ruina, 1983; Marone, 1998),

$$\mu(\boldsymbol{\xi},t) = \mu_0(\boldsymbol{\xi}) + a(\boldsymbol{\xi})\ln\left(\frac{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}{V_0(\boldsymbol{\xi})}\right) + b(\boldsymbol{\xi})\ln\left(\frac{\theta(\boldsymbol{\xi},t)V_0(\boldsymbol{\xi})}{L(\boldsymbol{\xi})}\right), \tag{23}$$

where μ_0, a, b, V_0, and L are empirical constants, and they can all vary across the fault.

A state evolution law is needed to constrain the state variable θ. The two most popular evolution laws used in dynamic rupture research are the aging law and the slip law.

$$\dot{\theta}(\boldsymbol{\xi},t) = 1 - \frac{\theta(\boldsymbol{\xi},t)|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}{L(\boldsymbol{\xi})}, \quad \text{Aging law} \tag{24}$$

$$\dot{\theta}(\boldsymbol{\xi},t) = -\frac{\theta(\boldsymbol{\xi},t)|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}{L(\boldsymbol{\xi})} \ln\left(\frac{\theta(\boldsymbol{\xi},t)|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}{L(\boldsymbol{\xi})}\right), \quad \text{Slip law} \tag{25}$$

Simpler than in the linear slip-weakening problem in Section 3.1, the rate-and-state problem does not have hybrid boundary conditions, and Equations (23) to (25) apply to all portions of the fault at all times. There are eight unknown variables to solve: $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)$ and $\boldsymbol{T}(\boldsymbol{\xi},t)$ (each with three components), as well as two friction law variables, $\mu(\boldsymbol{\xi},t)$ and $\theta(\boldsymbol{\xi},t)$. By combining (4), (5), (6), (23), (24) or (25), and the three equations in (3), we have eight equations to constrain the eight unknown variables.

The rate-and-state framework can be applied to simulate rupture cycles. Therefore, a constant stress loading is usually applied to the fault. Denoting $\eta(\boldsymbol{\xi})$ as the constant stressing rate at the location $\boldsymbol{\xi}$ on the fault, we can modify Equation (3) as follows to include the description of loading,

$$T_i(\boldsymbol{x},t) = T_i^0(\boldsymbol{x}) + \eta(\boldsymbol{\xi})t + \int_\Gamma \mathrm{d}S(\boldsymbol{\xi}) \int_0^t \mathrm{d}\tau \hat{K}_{ij}(\boldsymbol{x}, t-\tau; \boldsymbol{\xi}, 0)\Delta\dot{u}_j(\boldsymbol{\xi},\tau) \quad (x \in \Gamma) \tag{26}$$

We want to prove the following solution scaling. Let's say we have obtained the solution for $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)$, $\boldsymbol{T}(\boldsymbol{\xi},t)$, $\mu(\boldsymbol{\xi},t)$, and $\theta(\boldsymbol{\xi},t)$ for a given set of initial stress, initial state, stressing rate, and other constant rate-and-state parameters. If we linearly scale $\boldsymbol{T^0}$, η, V_0, and L by a factor of C, while keeping the other parameters unchanged, the solution for $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)$ and $\boldsymbol{T}(\boldsymbol{\xi},t)$ will be scaled by a factor of C, while $\mu(\boldsymbol{\xi},t)$ and $\theta(\boldsymbol{\xi},t)$ will remain unchanged.

Similar as in Section 3.1, we denote the solutions that satisfy the condition before 'scaling' as $\boldsymbol{T^a}(\boldsymbol{\xi},t)$, $\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$, $\mu^a(\boldsymbol{\xi},t)$, and $\theta^a(\boldsymbol{\xi},t)$. We can substitute these solutions into Equations (26), (4), (5), (6), and (23),

$$T_i^a(\boldsymbol{x},t) = T_i^0(\boldsymbol{x}) + \eta(\boldsymbol{\xi})t + \int_\Gamma \mathrm{d}S(\boldsymbol{\xi}) \int_0^t \mathrm{d}\tau \hat{K}_{ij}(\boldsymbol{x}, t-\tau; \boldsymbol{\xi}, 0)\Delta\dot{u}_j^a(\boldsymbol{\xi},\tau) \quad (x \in \Gamma) \tag{27}$$

$$\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t) \cdot \hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi}) = 0 \tag{28}$$

$$\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})=\mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\left(\frac{\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})\right) \tag{29}$$

$$\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})=\mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\left(\frac{\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})\right) \tag{30}$$

$$\mu^a(\boldsymbol{\xi},t)=\mu_0(\boldsymbol{\xi})+a(\boldsymbol{\xi})\ln\left(\frac{|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{V_0(\boldsymbol{\xi})}\right)+b(\boldsymbol{\xi})\ln\left(\frac{\theta^a(\boldsymbol{\xi},t)V_0(\boldsymbol{\xi})}{L(\boldsymbol{\xi})}\right) \tag{31}$$

249 The state evolution law could be either the aging or slip law,

$$\dot{\theta}^a(\boldsymbol{\xi},t)=1-\frac{\theta^a(\boldsymbol{\xi},t)|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{L(\boldsymbol{\xi})},\quad\text{Aging law} \tag{32}$$

$$\dot{\theta}^a(\boldsymbol{\xi},t)=-\frac{\theta^a(\boldsymbol{\xi},t)|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{L(\boldsymbol{\xi})}\ln\left(\frac{\theta^a(\boldsymbol{\xi},t)|\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{L(\boldsymbol{\xi})}\right),\quad\text{Slip law} \tag{33}$$

250 To prove the aforementioned statement, we scale $\boldsymbol{T^0}$, $\eta(\boldsymbol{\xi})$, V_0, and L by a factor
251 of C, and check whether $C\boldsymbol{T^a}(\boldsymbol{\xi},t)$, $C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$, $\mu^a(\boldsymbol{\xi},t)$, and $\theta^a(\boldsymbol{\xi},t)$ satisfy the new set
252 of equations. We can simply substitute the proposed solutions into the Equations (27)
253 to (33) and check,

$$CT_i^a(\boldsymbol{x},t)=CT_i^0(\boldsymbol{x})+C\eta(\boldsymbol{\xi})t+\int_\Gamma \mathrm{d}S(\boldsymbol{\xi})\int_0^t \mathrm{d}\tau\hat{K}_{ij}(\boldsymbol{x},t-\tau;\boldsymbol{\xi},0)C\Delta\dot{u}_j^a(\boldsymbol{\xi},\tau)\quad(x\in\Gamma) \tag{34}$$

$$C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})=0 \tag{35}$$

$$C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})=\mu(\boldsymbol{\xi},t)\Big(C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\left(\frac{C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})\right) \tag{36}$$

$$C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})=\mu(\boldsymbol{\xi},t)\Big(C\boldsymbol{T^a}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\left(\frac{C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)}{|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})\right) \tag{37}$$

$$\mu^a(\boldsymbol{\xi},t)=\mu_0(\boldsymbol{\xi})+a(\boldsymbol{\xi})\ln\left(\frac{|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{CV_0(\boldsymbol{\xi})}\right)+b(\boldsymbol{\xi})\ln\left(\frac{\theta^a(\boldsymbol{\xi},t)CV_0(\boldsymbol{\xi})}{CL(\boldsymbol{\xi})}\right) \tag{38}$$

254 The state evolution law could be either the aging or slip law,

$$\dot{\theta}^a(\boldsymbol{\xi},t)=1-\frac{\theta^a(\boldsymbol{\xi},t)|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{CL(\boldsymbol{\xi})},\quad\text{Aging law} \tag{39}$$

$$\dot{\theta}^a(\boldsymbol{\xi},t) = -\frac{\theta^a(\boldsymbol{\xi},t)|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{CL(\boldsymbol{\xi})}\ln\left(\frac{\theta^a(\boldsymbol{\xi},t)|C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)|}{CL(\boldsymbol{\xi})}\right), \quad \text{Slip law} \tag{40}$$

All the scaling factors C in the above equations will cancel out, and Equations (34) to (40) are identical to Equations (27) to (33). This means that the solutions $C\boldsymbol{T}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$, $C\Delta\dot{\boldsymbol{u}}^{\boldsymbol{a}}(\boldsymbol{\xi},t)$, $\mu^a(\boldsymbol{\xi},t)$, and $\theta^a(\boldsymbol{\xi},t)$ satisfy the new condition. Assuming the uniqueness of the solution, if the proposed solution is a solution, it is the only solution.

4 Numerical examples

In this section, we demonstrate the solution scaling behavior using two realistic numerical examples, one for dynamic rupture in a linear slip-weakening framework (Tarnowski et al., 2022), and one for fully dynamic rupture cycle models using the rate-and-state friction framework (Wu & Barbot, 2025).

4.1 Linear slip-weakening

Tarnowski et al. (2022) use linear slip-weakening dynamic rupture models to investigate potential rupture paths of earthquakes propagating through the structurally complex western San Gorgonio Pass, part of the San Andreas fault system in southern California (USA). Their simulations employ the 3D finite element code FaultMod (Barall, 2009), which has been extensively benchmarked in the dynamic rupture code validation exercises led by the Southern California Earthquake Center–U.S. Geological Survey (SCEC–USGS) dynamic rupture code validation group (Harris et al., 2018).

We run two dynamic rupture simulations using the complex fault system described by Tarnowski et al. (2022). One simulation replicates the experimental configuration of the evolved stress model that nucleates on the Garnet Hill segment from that study. The second simulation uses nearly identical configurations, except that the initial normal stress, initial shear stress, and slip-weakening distance are each scaled by an arbitrary factor of seven. We refer to the first simulation as the $C = 1$ case and the second as the $C = 7$ case.

The results of the two models are compared in Figure 1. The two slip maps have the same spatial distribution, with the $C = 7$ simulation showing exactly seven times the slip magnitude at every point on the fault compared to the $C = 1$ simulation (Fig-

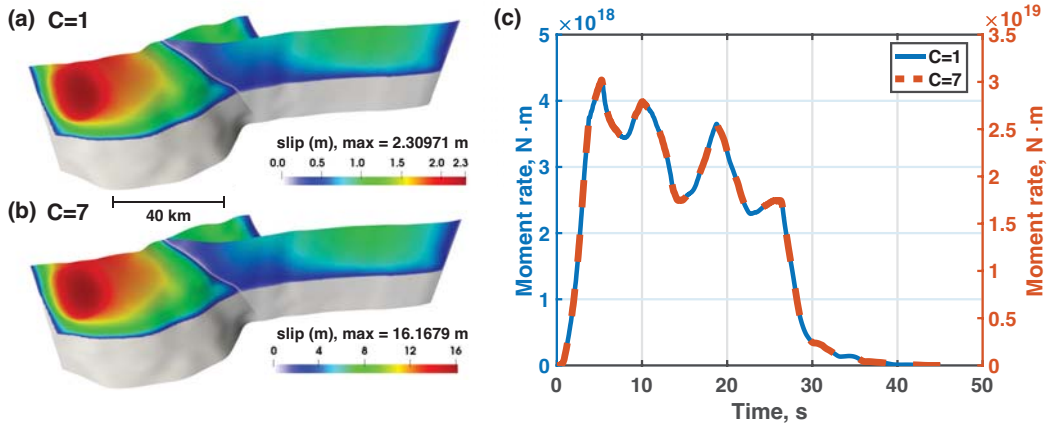


Figure 1. Comparison between two linear slip-weakening dynamic rupture simulations. **(a)** Final slip distributions of the dynamic rupture simulation with the same configuration as Model X in Tarnowski et al. (2022), denoted as the $C = 1$ simulation. **(b)** Final slip distributions of the dynamic rupture simulation with the initial stress $\boldsymbol{T}$ and D_0 scaled by seven from the $C = 1$ simulation, denoted as the $C = 7$ simulation. The final slip pattern remains the same, while the maximum slip and moment rate are seven times greater. **(c)** Moment rate functions of the $C = 1$ simulation (blue solid curve, y-axis to the left) and $C = 7$ simulation (orange dashed curve, y-axis to the right). The shapes of the moment rate function are the same, while the moment rate amplitude is seven times greater in the $C = 7$ simulation.

ure 1a, b). This result is consistent with the scaling properties explored in Section 3.1. We also present a comparison of the seismic moment rate functions for both cases (Figure 1c). The temporal evolutions of both ruptures are identical, except that the moment rate in the $C = 7$ simulation is scaled by a factor of seven.

4.2 Rate-and-state friction

(Wu & Barbot, 2025) use fully dynamic rupture cycle models to explain the evolution of the real area of contact during laboratory earthquake experiments. Their simulations employ a boundary element method, with the method details described in the paper. The model uses a physical rate-and-state friction framework, in which the constitutive relation between the friction coefficient, slip rate, and state variable takes a multiplicative form (Barbot, 2019), rather than the logarithmic additive form shown in Equa-

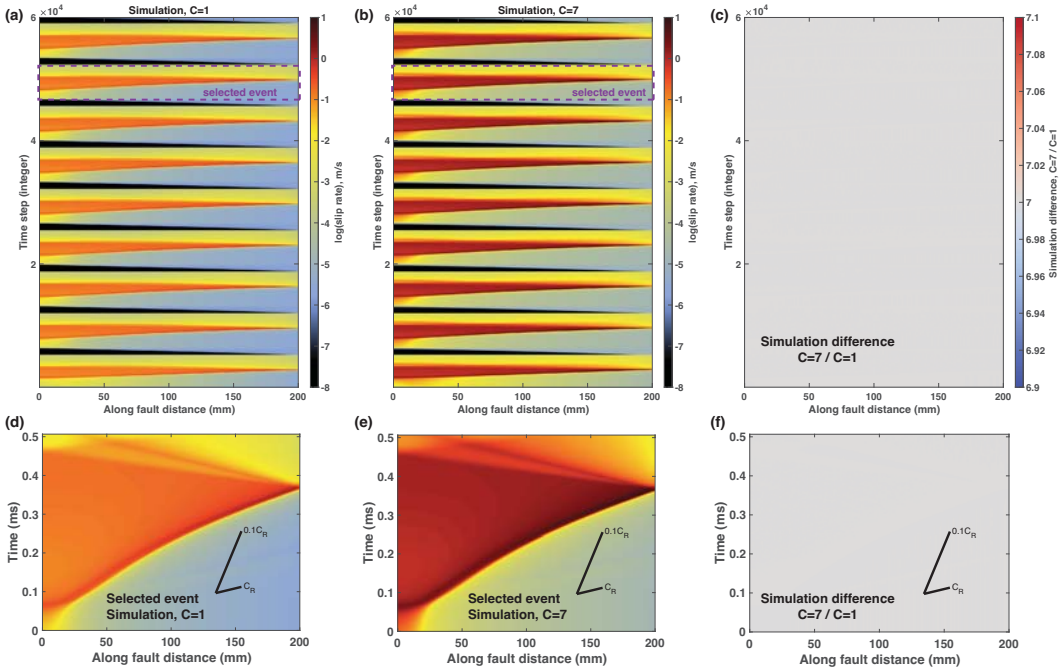


Figure 2. Comparison between two rate-and-state dynamic rupture cycle simulations. **(a)** Slip rate evolution versus simulation time step for the dynamic rupture cycle simulation with the same configuration as Simulation 2 in Wu and Barbot (2025), denoted as the $C = 1$ simulation. **(b)** Slip rate evolution versus simulation time step for the dynamic rupture cycle simulation with initial stress $\boldsymbol{T}$, stressing rate η, V_0, and L scaled by a factor of seven from the $C = 1$ simulation, denoted as the $C = 7$" simulation. **(c)** Difference in slip rate evolution versus simulation time step between the $C = 7$ simulation and the $C = 1$ simulation. The difference is exactly seven times. **(d)** Slip rate evolution versus actual time for one selected event in the $C = 1$ simulation. **(e)** Slip rate evolution versus actual time for one selected event in the $C = 7$ simulation. **(f)** Difference in slip rate evolution versus actual time between the $C = 7$ simulation and the $C = 1$ simulation. The difference is exactly seven times (grey).

tion (23),

$$\mu(\boldsymbol{\xi},t) = \mu_0(\boldsymbol{\xi})\left(\frac{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}{V_0(\boldsymbol{\xi})}\right)^{a(\boldsymbol{\xi})/\mu_0}\left(\frac{\theta(\boldsymbol{\xi},t)V_0(\boldsymbol{\xi})}{L(\boldsymbol{\xi})}\right)^{b(\boldsymbol{\xi})/\mu_0}. \tag{41}$$

Barbot (2019) suggests that the multiplicative form has a clear physical meaning and that the logarithmic additive form (Equation (23)) is an asymptotic approximation of the multiplicative form. Most importantly, one can trivially show that the solution scaling properties also apply to the multiplicative form by substituting Equation (23) with Equation (41) and following the same proof strategy as in Section 3.2.

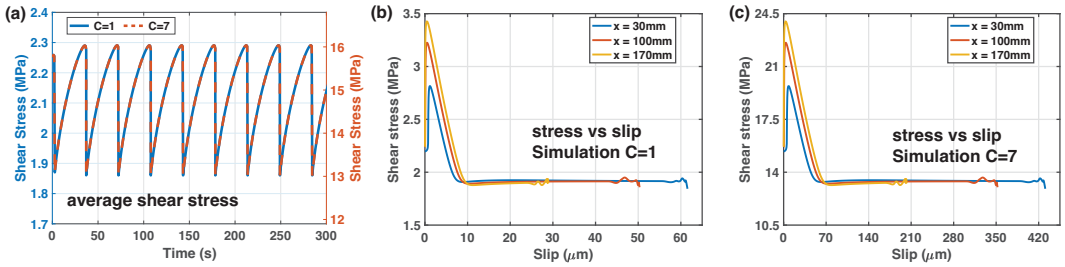


Figure 3. **(a)** Average shear stress on the fault versus actual time for the $C = 1$ simulation (blue solid curve, y-axis on the left) and the $C = 7$ simulation (orange dashed curve, y-axis on the right). The shapes of the average shear stress time series are the same, while the amplitude is seven times greater in the $C = 7$ simulation. **(b)** Shear stress versus slip relations at three different locations on the fault for a selected event in the $C = 1$ simulation. **(c)** Same as **(b)**, but for the selected event in the $C = 7$ simulation.

We conducted two fully dynamic rupture cycle simulations using the rate-and-state framework, adapting the model configurations from Wu and Barbot (2025). The first simulation has an identical configuration to Simulation 2 in Wu and Barbot (2025), which we denote as the $C = 1$ simulation. The second simulation is nearly identical to the $C = 1$ simulation, except that the initial shear stress, initial normal stress, stressing rate η, rate-and-state reference velocity V_0, and state evolution characteristic distance L are each scaled by a factor of seven. We denote this second simulation as the $C = 7$ simulation.

We ran both simulations for 60,000 time steps. The two simulations exhibit identical cycle patterns with nine seismic events (Figure 2a,b); the coseismic rupture patterns are the same between the two simulations (Figure 2d,e). Note that we use an adaptive time-stepping scheme; therefore, most time steps are concentrated in the coseismic stage (Wu & Barbot, 2025). We divide the slip rate from the $C = 7$ simulation by that of the $C = 1$ simulation, and the ratio is seven at all fault locations and at every time step (Figure 2c,f).

We compare the stress and slip evolutions between the two simulations. The relative stress drop and absolute stress level are seven times greater in the $C = 7$ simulation than in the $C = 1$ simulation (Figure 3a), while the shapes of the average shear stress time series are the same. In addition, the shear stress versus slip relations are self-

similar between the two simulations; both the effective slip-weakening distance and the final slip in the $C = 7$ simulation are seven times greater than in the $C = 1$ simulation. As expected, these numerical results are consistent with the solution scaling analysis in Section 3.2.

5 Discussion

5.1 Relation to the general solution scaling in linear elasticity

The dynamic rupture problem belongs to the category of linearized elasticity problems, which are well known to have solution scaling properties with certain simple boundary conditions (Slaughter, 2002). For example, consider a problem where a body occupies the region $\mathcal{R}$ with boundary $\partial\mathcal{R}$. Within the elastic body, the displacement vector $u_i(\boldsymbol{x}, t)$, strain tensor $\varepsilon_{ij}(\boldsymbol{x}, t)$, and stress tensor $\sigma_{ij}(\boldsymbol{x}, t)$ satisfy the equations of motion, the strain-displacement equation, and Hooke's law as the constitutive equations,

$$\frac{\partial \sigma_{ji}}{\partial x_j} + f_i = \rho \frac{\partial^2 u_i}{\partial t^2}, \tag{42}$$

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \tag{43}$$

$$\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl}, \tag{44}$$

f_i is the body force, which is often neglected in dynamic rupture problems. ρ is medium density. C_{ijkl} is the fourth-order stiffness tensor. In an isotropic medium, Equation (44) can be simplified as follows:

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij}, \tag{45}$$

δ_{ij} is the Kronecker delta. λ is Lamé's first parameter, and μ is the shear modulus, both of which are scalar quantities and can vary in space.

Considering that the boundary satisfies either a displacement boundary condition $\tilde{u}_i$ at the portion $\partial\mathcal{R}^u$ or a traction boundary condition $\tilde{T}_i$ at the portion $\partial\mathcal{R}^t$,

$$u_i(\boldsymbol{x}, t) = \tilde{u}_i(\boldsymbol{x}), \quad \boldsymbol{x} \in \partial\mathcal{R}^u \tag{46}$$

$$\sigma_{ij}(\boldsymbol{x}, t) n_j = \tilde{T}_i(\boldsymbol{x}), \quad \boldsymbol{x} \in \partial\mathcal{R}^t, \tag{47}$$

where n_i is the normal vector at the boundary.

We can identify scaling relations from these equations through simple algebra. Let's denote the solution to a given boundary condition, body force, and elastic medium properties as $u_i^{(1)}$, $\varepsilon_{ij}^{(1)}$, and $\sigma_{ij}^{(1)}$, with a superscript (1). We can scale the spatial length by C_x, the stress tensor by C_σ, and the displacement vector by C_u, and denote the new fields with a superscript (2),

$$x_i^{(2)} = \frac{x_i^{(1)}}{C_x}, \quad \sigma_{ij}^{(1)} = \frac{\sigma_{ij}^{(1)}}{C_\sigma}, \quad u_i^{(2)} = \frac{u_i^{(1)}}{C_u}, \quad \varepsilon_{ij}^{(2)} = \frac{C_x}{C_u}\varepsilon_{ij}^{(1)}. \tag{48}$$

The new field (2) satisfies a new set of boundary conditions, body forces, and elastic medium properties. The relations between the new and old parameters are:

$$f_i^{(2)} = \frac{C_x}{C_\sigma} f_i^{(1)}, \quad \rho^{(2)} = \frac{C_u C_x}{C_\sigma}\rho^{(1)}, \quad C_{ijkl}^{(2)} = \frac{C_u}{C_x C_\sigma} C_{ijkl}^{(1)}, \quad \tilde{u}_i^{(2)} = \frac{\tilde{u}_i^{(1)}}{C_u}, \quad \tilde{T}_i^{(2)} = \frac{\tilde{T}_i^{(1)}}{C_\sigma}. \tag{49}$$

The solution scaling relations discussed above hold for simple displacement and traction boundary conditions. However, they do not apply to dynamic rupture problems due to complications in the associated boundary conditions. In a dynamic rupture setting, faults must be treated as internal boundaries (Aki & Richards, 2002; Tada, 2009) and fault rheology acts as an internal boundary condition: displacement is discontinuous across the fault, while the stress tensor remains continuous. As described in Section 2, the fault boundary condition is typically expressed as a constitutive relation between traction and slip (i.e., the discontinuity in displacement across the fault). Additional complexities may arise. In the linear slip-weakening framework, a location on the fault may be subject to either a simple zero-slip boundary condition or a linear slip-weakening stress profile, depending on whether the shear traction has ever exceeded the yielding strength at that location (Section 3.1). In the rate-and-state framework, the friction coefficient is often defined through a set of ordinary differential equations with respect to time, involving an additional state variable (Section 3.2).

In Section 3, we show that solution scaling may exist for certain dynamic rupture problems. In fact, the parameter scaling in dynamic rupture problems represents a more restrictive form of the scaling relations found in linear elastic problems with simple boundary conditions. Using Equations (48) and (49), one may independently scale the spatial length by a factor C_x, the displacement by C_u, and the stress by C_σ, and consequently obtain new fields that satisfy a new set of boundary conditions. In contrast, for dynamic rupture problems, the parameter scaling requires $C_\sigma = C_u$ and $C_x = 1$. This stricter requirement arises because the introduction of a friction law introduces coupling between

stress and slip, preventing stress and displacement from being scaled independently. Additionally, the friction law introduces a new length scale (e.g., D_0 in the linear slip-weakening framework), which further restricts independent scaling of the spatial coordinates x_i.

Lastly, we note that it is possible to deduce solution scaling for dynamic rupture problems starting from Equations (42) to (47), rather than using boundary integral equations. We have chosen the boundary integral formulation because it offers a more straightforward mathematical framework for handling fault boundary conditions. In addition, a solution scaling relation for the spatial coordinates x_i likely exists for certain special cases of dynamic rupture problems: for example, rupture on a planar fault embedded in a three-dimensional, isotropic, and homogeneous full space (Fukuyama & Madariaga, 1998). To derive such solution scaling relations using boundary integral equations, one would need to explicitly express the integral kernel $\hat{K}_{ij}$ in Equation (3).

5.2 Existence and uniqueness of the solution

We note that the proof we provided here is not entirely rigorous because we assume the existence and uniqueness of the solution to the described dynamic rupture problem. Indeed, the partial differential equations that describe the elastodynamic system have a unique solution when given the initial and boundary conditions, provided that the boundary condition is either a displacement or a traction boundary condition, applied in a mutually exclusive way on disjoint parts of the boundary (Slaughter, 2002). However, the boundary conditions most commonly used in dynamic rupture models describe the relation between shear and normal stress and its evolution. In the slip-weakening case, the type of boundary condition can even change dynamically (Equation (7)). Therefore, it is not guaranteed that such a problem has a unique solution, and we are making a significant assumption in our derivation. Nevertheless, we believe it is reasonable to assume that there should be a unique solution to a physics-based problem. Further analysis can be conducted to test this assumption.

5.3 Applying the analysis to other fault rheology formulations

The boundary integral equation approach presented in Section 3 is general; therefore, it could also be used to derive solution scaling relations for other fault rheology formulations. Specifically, we can write out the specific formulation of $\mu(\boldsymbol{\xi}, t)$ and combine

it with Equations (3) to (6). Then, we may assume a solution is known, scale it by a constant factor C, and identify how to modify the initial model parameters such that all the scaling factors C cancel out in the equations. Necessary modifications might be needed, such as the partitioning of fault in Section 3.1 or the introduction of loading rate in 3.2.

It is important to note that this approach requires two conditions. First, the medium properties remain unchanged, and the boundary conditions on the model domain boundary are homogeneous (i.e., zero displacement or traction). This condition ensures that the integration kernel $\hat{K}_{ij}$ in Equation (3) does not change with different initial stress and fault rheology parameters. Second, there is a solution to the specific equations, and the solution is unique (Section 5.2).

As examples, we discuss below two additional formulations that are occasionally used in dynamic rupture studies: linear time-weakening friction and composite formulations with different slip-rate dependencies at different slip rates. We only show the analytical proof; numerical examination will be a subject for future study.

5.3.1 Linear time-weakening friction

Linear time-weakening fault rheology is sometimes used in dynamic rupture models as an effective parameterization to capture the effects of off-fault inelastic processes (e.g., Andrews, 2004, 2005). In some cases, linear time-weakening has been used for the nucleation process while the overall rupture outside the nucleation patch is governed by linear slip-weakening (e.g., Bizzarri et al., 2001; Bizzarri, 2010; Hu et al., 2017).

Similar to the linear slip-weakening case, a linear time-weakening dynamic rupture problem involves two types of boundary conditions on different portions of the fault. Before rupturing, a fault location satisfies a Dirichlet boundary condition with slip $\Delta \boldsymbol{u} = 0$; after rupturing, the friction coefficient μ at this fault location decreases linearly with time. As in Section 3.1, we denote the subset of the fault that has ruptured at time t as $\Gamma_A(t)$, and the complementary subset that has not ruptured as $\Gamma_B(t)$. They can be defined as,

$$\boldsymbol{\xi} \in \begin{cases} \Gamma_A(t), \text{ if } |\boldsymbol{T}(\boldsymbol{\xi},t) - \big(\boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\big)\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| = \mu_y(\boldsymbol{\xi})|\boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})| \text{ or } |\Delta\boldsymbol{u}(\boldsymbol{\xi},\tau)| > 0 \\ \Gamma_B(t), \text{ otherwise.} \end{cases} \tag{50}$$

The fault boundary conditions, Equations (5) and (6), can be revised as,

$$\begin{cases} \boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\left(\dfrac{\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi})\right), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ \Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{1}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{51}$$

$$\begin{cases} \boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = \mu(\boldsymbol{\xi},t)\Big(\boldsymbol{T}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{3}}(\boldsymbol{\xi})\Big)\left(\dfrac{\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)}{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)|}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi})\right), & \text{if } \boldsymbol{\xi}\in\Gamma_A(t) \\ \Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi},t)\cdot\hat{\boldsymbol{n}}_{\boldsymbol{2}}(\boldsymbol{\xi}) = 0, & \text{if } \boldsymbol{\xi}\in\Gamma_B(t) \end{cases} \tag{52}$$

To write out the formulation of linear time-weakening rheology, we define a new variable on the fault, $t_r(\boldsymbol{\xi})$, which records the time when a fault location first ruptures. The rupture time function, $t_r(\boldsymbol{\xi})$, is not an independent variable; it depends on the slip solution, $\Delta\boldsymbol{u}(\boldsymbol{\xi},t)$, although the mapping is difficult to express analytically. For fault portions that never rupture, we may need to assign an arbitrary value to $t_r(\boldsymbol{\xi})$.

We may then write the expression for $\mu(\boldsymbol{\xi},t)$ in the linear time-weakening formulation using $t_r(\boldsymbol{\xi})$. After rupturing, the friction coefficient μ decreases linearly from μ_y to μ_d over a critical weakening time $t_0(\boldsymbol{\xi})$,

$$\mu(\boldsymbol{\xi},t) = \begin{cases} \mu_y(\boldsymbol{\xi})\left(1-\dfrac{t-t_r(\boldsymbol{\xi})}{t_0(\boldsymbol{\xi})}\right)+\mu_d(\boldsymbol{\xi}), & t-t_r(\boldsymbol{\xi})<t_0(\boldsymbol{\xi}) \\ \mu_d(\boldsymbol{\xi}), & t-t_r(\boldsymbol{\xi})\geq t_0(\boldsymbol{\xi}) \end{cases} \quad (\boldsymbol{\xi}\in\Gamma_A(t)), \tag{53}$$

We note that the rupture time variable $t_r(\boldsymbol{\xi})$ should remain unchanged when the slip solution $\Delta\boldsymbol{u}(\boldsymbol{\xi},\tau)$ is scaled linearly. Therefore, following the same analysis approach as in Equations (11) to (22), we can identify the solution scaling relation for linear time-weakening rheology as follows: If we linearly scale $\boldsymbol{T^0}$ by a factor of C, while keeping the other parameters unchanged, the solutions $\boldsymbol{T}(\boldsymbol{\xi},t)$ and $\Delta\boldsymbol{u}(\boldsymbol{\xi},t)$ will also be scaled by a factor of C. Note that we do not scale the critical weakening time $t_0(\boldsymbol{\xi})$.

5.3.2 Composite rheology with different slip-rate dependencies at different slip rates

It has been recognized that friction strength may drastically reduce at high slip rates, and the corresponding physical mechanisms, such as flash heating (e.g., Tsutsumi & Shimamoto, 1997; Di Toro et al., 2004; Beeler et al., 2008; Goldsby & Tullis, 2011) and thermal pressurization (e.g., Acosta et al., 2018; Badt et al., 2020), may differ from those that control friction strength at low slip rates. These observations have led some researchers to formulate effective composite fault rheologies with different slip rate dependencies at

different slip rates for dynamic rupture models (e.g., Ampuero & Ben-Zion, 2008; Rojas et al., 2009; Gabriel et al., 2012).

We may investigate the solution scaling characteristics of these dynamic rupture models using a similar approach as in the analysis for the empirical rate-and-state friction law (Section 3.2). We may assume a general formulation of the friction coefficient $\mu(\boldsymbol{\xi}, t)$,

$$\mu(\boldsymbol{\xi}, t) = \begin{cases} f_1\left(\dfrac{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)|}{V_1(\boldsymbol{\xi})}\right), & |\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)| < V^*(\boldsymbol{\xi}) \\ f_2\left(\dfrac{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)|}{V_2(\boldsymbol{\xi})}\right), & |\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)| \geq V^*(\boldsymbol{\xi}) \end{cases} \qquad (\boldsymbol{\xi} \in \Gamma), \tag{54}$$

where f_1 and f_2 are two general functions relating slip rate to friction coefficient. V_1, V_2, and V^* are three model parameters that could vary on fault.

Following the same analysis approach as in Equations (27) to (40), we can identify the solution scaling relation for the above friction law (Equation (54)) as follows: If we linearly scale the initial stress $\boldsymbol{T^0}$, stressing rate η, and friction parameters V_1, V_2, and V^* by a factor of C, while keeping the other parameters in the functions f_1 and f_2 unchanged, then the solutions for $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$ and $\boldsymbol{T}(\boldsymbol{\xi}, t)$ will be scaled by a factor of C, while the friction coefficient $\mu(\boldsymbol{\xi}, t)$ will remain unchanged.

An alternative type of composite formulation is to describe a composite evolution law formulation switching at a transition slip rate V^* (e.g., Rojas et al., 2009). A general formulation of the friction coefficient $\mu(\boldsymbol{\xi}, t)$ and state variable ψ can be written as,

$$\mu(\boldsymbol{\xi}, t) = f_0\left(\frac{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)|}{V_0(\boldsymbol{\xi})}, \psi(\boldsymbol{\xi}, t)\right) \qquad (\boldsymbol{\xi} \in \Gamma), \tag{55}$$

and the state evolution law can be written as,

$$\dot{\psi}(\boldsymbol{\xi}, t) = \begin{cases} f_1\left(\dfrac{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)|}{V_1(\boldsymbol{\xi})}, \psi(\boldsymbol{\xi}, t)\right), & |\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)| < V^*(\boldsymbol{\xi}) \\ f_2\left(\dfrac{|\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)|}{V_2(\boldsymbol{\xi})}, \psi(\boldsymbol{\xi}, t)\right), & |\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)| \geq V^*(\boldsymbol{\xi}) \end{cases} \qquad (\boldsymbol{\xi} \in \Gamma), \tag{56}$$

We may again identify the solution scaling relation using the same analysis approach as in Equations (27) to (40): If we linearly scale the initial stress $\boldsymbol{T^0}$, stressing rate η, and friction parameters V_0, V_1, V_2, and V^* by a factor of C, while keeping the other parameters in the functions f_0, f_1, and f_2 unchanged, then the solutions for $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$ and $\boldsymbol{T}(\boldsymbol{\xi}, t)$ will be scaled by a factor of C, while the friction coefficient $\mu(\boldsymbol{\xi}, t)$ and the state variable $\psi(\boldsymbol{\xi}, t)$ will remain unchanged.

5.4 Solution scaling for dynamic rupture models beyond linear elasticity

In this study, we limited our scope to the solution scaling relation for linear elastic dynamic rupture problems. Our analytical approach does not address whether general solution scaling relations exist beyond the constraints of linear elasticity: for example, in models that incorporate off-fault inelastic yielding (e.g., Gabriel et al., 2013; S. Xu & Ben-Zion, 2013; Y. Wang & Day, 2020), or in models that include near-fault hydrothermal processes (e.g., Andrews, 2002; Rice, 2006; Noda et al., 2009).

Introducing additional physical processes into dynamic rupture models would introduce new length or time scales, as well as new couplings with elastic stresses. These additional coupling relations may not be compatible with the intrinsic self-similarity present in linear elasticity problems. Therefore, one may qualitatively suspect that a general solution scaling is less likely to exist for dynamic rupture problems beyond linear elasticity, although it may still exist for certain specific configurations.

5.5 Effects of the choice of numerical scheme

We note that numerical solutions to dynamic rupture problems are always affected by the specific numerical scheme chosen. For example, spurious high-frequency numerical noise can originate from insufficient resolution of physical processes, such as elastodynamic interactions or the breakdown process near the rupture front, and may contaminate the solution across all space and time scales (e.g., Day et al., 2005). In other methods, such as Discontinuous Galerkin approaches, higher frequencies may be damped due to limited resolution (Pelties et al., 2014).

In addition, some numerical schemes may artificially introduce extra length or time scales. For instance, a dynamic rupture cycle model may use an artificial threshold to determine whether to apply a quasi-dynamic or fully dynamic solver (Erickson et al., 2023). These artificial scales may cause the simulation to violate the solution scaling relation. Therefore, when applying the solution scaling predicted by the present analytical analysis, care must be taken to ensure that numerical realizations are not strongly affected by the choice of numerical scheme.

5.6 Differences from the earthquake source scaling relations based on constant stress drop model

It is worth noting that the solution scaling discussed in this paper is different from the earthquake source scaling relations derived from the constant stress drop model (e.g., Kanamori & Anderson, 1975; Kanamori & Brodsky, 2004), although both topics are fundamentally related to the linearity of elasticity.

Our theoretical results are developed to justify common practices in dynamic rupture modeling and to help modelers scale dynamic rupture solutions for benchmarking and sensitivity analysis. The scaling relations derived in this study confirm that linear elastic dynamic rupture solutions remain invariant under proportional changes in stress and key frictional parameters, provided the model assumptions are met. This has practical value for efficiently exploring model parameter space and non-dimensionalizing results.

Earthquake source scaling relations, on the other hand, are based on observed earthquake source parameters, such as seismic moment, rupture area, slip, duration, and radiated energy. The empirical relations among these observed parameters across different earthquake sizes suggest that earthquakes of various magnitudes may share a similar characteristic stress drop of approximately ~ 10 MPa, though with uncertainty spanning orders of magnitude (e.g., Kanamori & Brodsky, 2004; Allmann & Shearer, 2009; Abercrombie, 2021).

6 Conclusion

In this paper, we demonstrate that when scaling the initial stress and other fault boundary conditions in a dynamic rupture model in a certain manner, the new solution will differ from the original solution only by a scaling factor. We consider four types of fault rheology: linear slip-weakening, rate-and-state friction, linear time-weakening, and general composite rheology with different slip-rate dependencies at different slip rates.

For linear slip-weakening rheology (Section 3.1), we show that if the initial on-fault traction $\boldsymbol{T^0}$ and characteristic slip-weakening distance D_0 are linearly scaled by a factor of C, while keeping the other parameters unchanged, the solutions $\boldsymbol{T}(\boldsymbol{\xi}, t)$ and $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$ will also be scaled by a factor of C.

For the empirical rate-and-state friction rheology (Section 3.2), we show that if the initial on-fault traction $\boldsymbol{T^0}$, stressing rate η, rate-and-state parameter V_0, and rate-and-state characteristic slip distance L are linearly scaled by a factor of C, while keeping the other parameters unchanged, the solutions for $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$ and $\boldsymbol{T}(\boldsymbol{\xi}, t)$ will be scaled by a factor of C, while $\mu(\boldsymbol{\xi}, t)$ and $\theta(\boldsymbol{\xi}, t)$ remain unchanged.

For linear time-weakening rheology (Section 5.3.1), we show that if the initial on-fault traction $\boldsymbol{T^0}$ is linearly scaled by a factor of C, while keeping the other parameters unchanged, the solutions $\boldsymbol{T}(\boldsymbol{\xi}, t)$ and $\Delta\dot{\boldsymbol{u}}(\boldsymbol{\xi}, t)$ will also be scaled by a factor of C. Note that we do not scale the critical weakening time $t_0(\boldsymbol{\xi})$.

For a general composite rheology with different slip-rate dependencies at different slip rates, the scaling relation can vary depending on the specific formulation. However, it can still be derived using the same approach as for the other fault rheologies (Section 5.3.2).

Since we formulated the dynamic rupture problem using a set of general boundary integral equations, the solution scaling relations are applicable to arbitrary free-surface shapes, medium velocity structures, and fault geometries, and should hold regardless of the specific numerical method employed, such as boundary element, finite element, or finite difference methods. Dynamic rupture models play an increasingly important role in earthquake physics research and seismic hazard assessment. We hope the scaling relations presented in this study will assist future dynamic rupture modelers and facilitate more systematic and efficient use of dynamic rupture models across a wide range of earthquake studies.

Open Research Section

This study is entirely theoretical and does not rely on new observational data. All information required to evaluate the conclusions is contained within the paper and its references. The model formulations and derivations are described in detail in the manuscript. No new data were generated, and no data or code are archived.

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